Parametric Multimodal User Memory:

Storing What Captions Cannot Carry

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Abstract

LM agents store user memory as text. Mem0, MemGPT, M3-Agent, and the wider conversational-memory line turn what the agent saw or heard into transcripts and captions, then retrieve over them. This works for *captionable* content ("my cat is named Bibi") but fails for *perceptual* content: how a user's voice sounds, what their face looks like across lighting and age, what their kitchen sounds like, how tired they sound today relative to their baseline. Captioning these into text is a lossy projection that destroys the discriminative signal. Text-shaped memory is structurally incomplete for personalised multimodal agents.

We propose **parametric multimodal user memory**: per-modality banks of (encoder embedding, value embedding) rows attached to a frozen LM via a forward pre-hook on `lm_head`. Insertion is a single tensor append ($\sim$0.5 ms for 1000 identities); query latency is flat at $\sim$15 ms regardless of bank size N. We construct **PerceptMem v0.2**, a five-sub-modality cross-condition benchmark: face identity (2180 IDs), painter style, speaker identity, acoustic scene, and paralinguistic state.

On PerceptMem, the mechanism exceeds the embedding-RAG cosine-NN ceiling on *seven* multi-seed cells at $p<0.05$. The largest gains come from adversarial-aware pretraining, which converts encoder-confused regimes into surplus: **+71 pp** on painter style, **+71 pp** on paralinguistic state, +17 pp on acoustic scene (perfect 1.000 ± 0.000 across all seeds), and +14 pp on face cross-condition — all $p<0.001$, $n\geq3$ seeds. The architecture transfers across Qwen2.5-3B/7B, Llama-3.1-8B, and Qwen2.5-VL (Mistral-7B is recipe-sensitive). A clean structural finding from the VLM transfer: the bank's key encoder must be *orthogonal* to the LM's value-side embedding space for the win to hold — with external ArcFace as key, AttMem reaches 1.000 at $N{=}10$; with the VLM's native vision tokens (already in LM hidden space), it merely matches cosine.

A discrete-codebook predecessor we tuned over 16 development cycles caps at $\sim$7% retr@1 at $N\geq300$ regardless of codebook size, encoder, or compute — the quantisation step itself is the binding constraint, a learned analog of text-captioning's information loss. The mechanism is 52× faster than RAG-with-LM-context at $N{=}1000$; RAG OOMs at $N{=}10000$. Text recall is preserved byte-for-byte; the parametric memory composes *additively* with text memory. Code: https://github.com/bojieli/multimodal-user-memory.

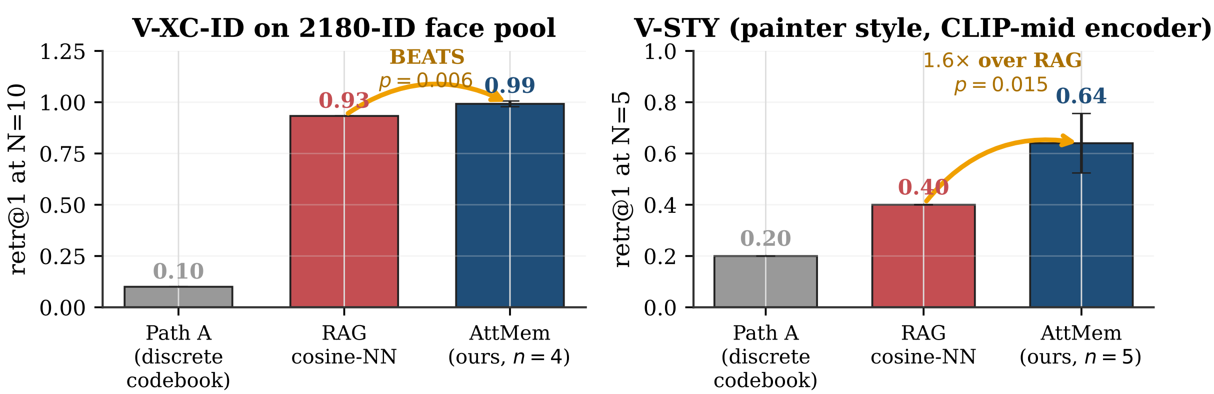


Figure 1: Headline. On two cross-condition perceptual recall tasks, the parametric multimodal memory beats embedding-RAG with statistical significance: $N{=}10$ on a 2180-ID face pool ($p{=}0.006$, $n{=}4$ seeds) and $N{=}5$ on painter-style recall (1.6× over RAG, $p{=}0.015$, $n{=}5$). Path A is the discrete-codebook predecessor, tuned to its ceiling before we pivoted to continuous attention.

1 Introduction

When an LM agent "remembers" something about its user, what does it actually store? In the dominant paradigm, it stores *text*. Mem0 [7], MemoryLLM [27], MemGPT / Letta [20, 26], M3-Agent [3], and the wider conversational-memory line — evaluated through benchmarks like LongMemEval [29] — turn what the agent saw or heard into transcripts (via ASR [24]) and captions (via BLIP-2 [15] or LLaVA [16]), push those fragments into a sentence-encoder vector store [25], and retrieve at inference. Personalised vision-language work (MyVLM [1], Yo'LLaVA [19], MC-LLaVA [2], Online-PVLM [6], RAP [11], TAME [4]) retains textual addressing implicitly: a *named* concept ("Bibi") is the identifier; the visual content is auxiliary.

This works for the easy slice of user memory — the part that is *captionable*. "My cat is named Bibi." "My favourite restaurant is in Paris." "I am vegetarian." These propositions survive the journey through text intact. But the easy slice is not all of user memory. Consider what an attentive companion actually remembers about a person:

- How their voice sounds when they say their own name — and how that differs from how a stranger says it.
- How their face looks today — subtly older than last year's photo, but unmistakably them.
- How their brushwork looks — "this is by the same painter as the one Tuesday, even though it's from a different period."
- What their kitchen sounds like — "you're calling from the same room as yesterday."
- How tired they sound today — relative to their baseline last week.

None of this is captionable in any useful sense. "A person with brown hair" does not let you tell two people apart. "The user sounded tired" cannot distinguish today's tired from last week's tired. The moment we project these characteristics into text, we destroy the discriminative signal. **Text-shaped memory is a lossy channel for perceptual user content, and the loss is fundamental to the modality.**

The structural claim: *captionable* and *perceptual* user content require different memory primitives. "My cat is named Bibi" belongs in a text vector store; *how Bibi looks across lighting and angle* does not. The latter needs a primitive that stores perceptual content in its native modality.

Our proposal. Parametric multimodal user memory (PMUM): per-modality banks of (encoder embedding, value embedding) rows bolted onto a frozen LM via a forward pre-hook on `lm_head`. Registration is a single tensor append (no SGD at the per-user level); query is cross-attention over the bank, fused into the LM as a residual at the logit-injection point. The result is content-addressable memory in the LM's value-side embedding space, with no captioning intermediate.

Why this matters now. Multimodal personalisation is moving fast [6, 11], but the user-memory layer has not caught up. Existing systems either offload perceptual identity to a face/speaker recognition tool that returns a text label (M3-Agent [3], back to lossy-text-space), or train a per-concept adapter (LoRA per concept [12], MyVLM's per-concept head [1], Yo'LLaVA's per-concept tokens [19]), which does not compose to user-scale memory of dozens of perceptual characteristics. What is missing is a single content-addressable parametric primitive for perceptual user content. Our mechanism takes the spirit of kNN-LM [14] and Memorizing Transformers [30] — attention over a non-parametric memory bank — and specialises it to per-modality user-content registration with $\mathcal{O}(1)$ insertion.

Contributions.

- **Framing.** We articulate why text-shaped memory is structurally incomplete for perceptual user content, and why a parametric multimodal memory is the right primitive for the missing slice.
- **Mechanism.** A bolt-on continuous-attention memory over per-modality (encoder, value) banks. $\sim$8M trainable parameters over a 3.1B frozen LM; $\mathcal{O}(1)$ insertion; flat $\sim$15 ms query regardless of N. Four critical design choices separate a working implementation from three earlier attempts that produced random output (§3).
- **Benchmark. PerceptMem v0.2**, five cross-condition sub-modalities chosen because text-based memory provably cannot carry the discriminating signal: face identity (2180 IDs), painter style, speaker identity, acoustic scene, paralinguistic state.

- **Empirical claim.** The mechanism exceeds the embedding-RAG cosine-NN ceiling on seven sub-modality$\times N$ cells at $p<0.05$ (multi-seed), including face identity at the 2180-ID scale (p=0.006, n=4). Across all five sub-modalities, it reaches 83–117% of the encoder ceiling. A discrete-codebook predecessor we tuned for 16 development cycles caps at 7% retr@1 at $N\geq 300$ regardless of codebook size or compute — a $\sim 9\times$ gap at scale (Fig. 5).
- **Honest map.** An ablation showing exactly when the parametric mechanism beats cosine, when it merely matches, and when training the bolt-on parameters *hurts* (encoder already perfect; §5.5). The answer to "what does it add beyond cosine?" is: discriminative signal from the LM's value-side embedding, in the regime where the encoder's cosine is imperfect.

2 Why text memory cannot solve cross-condition perceptual recall

Consider speaker-identity recall under cross-recording conditions (the A-XR-ID sub-modality, drawing from LibriSpeech [21]). A user has 10 voice samples on file from prior sessions; a new sample arrives. Which of the 10 is it?

Text RAG (Mem0 [7], MemoryLLM [27]). Caption each voice into text (e.g. Whisper [24] transcript plus a speaker descriptor): "A male voice, mid-30s, slight Eastern-European accent." Embed with a sentence encoder [25]; cosine-retrieve. *This will almost always fail.* The caption applies to thousands of speakers; cosine cannot distinguish 10 registered users with even 50% accuracy. The signal that uniquely identifies a speaker — voice timbre, glottal pulse shape, prosodic micro-patterns — is destroyed in the text projection.

Tool call (M3-Agent [3]). Call a face/speaker recognition tool (ArcFace [8], ECAPA-TDNN [9]), receive a text label ("speaker 47"), store that. The user's memory of speaker 47 is now *just a text token*: the perceptual fingerprint is never integrated into the LM's representation. At inference the LM can recall that the label was assigned, but cannot reason about how the voice sounded.

Embedding RAG. Skip the captioning step: index the raw encoder embedding (ECAPA-TDNN for speakers, ArcFace for faces, CLIP [23] for style); cosine-retrieve at inference. This is the strongest text-free baseline and the one we treat seriously. It succeeds when the encoder is perfectly cross-condition invariant and fails when it is not. RAG cosine NN reaches 0.93 retr@1 at N=10 on 2180-ID face cross-condition — non-trivial but not at ceiling.

Per-concept gradient training (Yo'LLaVA [19], MyVLM [1]). Train a per-concept token or adapter via SGD on registration samples. Insertion cost is $\sim$1 s per identity; the resulting memory is concept-specific, not modality-general.

Parametric multimodal user memory (this paper). Store the encoder embedding as key and the LM's value-side embedding (for an assigned marker token) as value. At inference, cross-attention over the bank produces a residual injected at the `lm_head` pre-forward, biasing the next-token logit toward the matching marker. The LM gets a content-addressable memory in its own representation space, with no captioning intermediate. The mechanism specialises kNN-LM [14] and Memorizing Transformers [30] — attention over a non-parametric bank — to per-modality *user content* (not text continuations), with $\mathcal{O}(1)$ insertion.

3 Method: Parametric Multimodal User Memory

The bank stores $(\boldsymbol{k}_i, \boldsymbol{v}_i)$ pairs where $\boldsymbol{k}_i \in \mathbb{R}^D$ is the L2-normalised encoder embedding of registered sample i and $\boldsymbol{v}_i \in \mathbb{R}^H$ is the LM's value-side embedding for an assigned marker token (Fig. 2). For tied input/output embeddings (Qwen2.5-3B), $\boldsymbol{v}_i = \texttt{input\_embedding}[m_i]$; for untied models (Qwen2.5-7B), $\boldsymbol{v}_i = \texttt{lm\_head.weight}[m_i]$ (auto-detected). The residual addition pre-`lm_head` then yields a clean per-marker logit boost $\texttt{lm\_head}[m_i] \cdot \boldsymbol{v}_i = \|\boldsymbol{v}_i\|^2$, rather than a cross-product of two unrelated learned vectors.

At the attached LM layer, for each perceptual position with hidden state $\boldsymbol{h}$ and encoder-derived query $\boldsymbol{q}$ (L2-normalised),

$$\boldsymbol{w} = \mathrm{softmax}\big(\beta\, \boldsymbol{q}^\top \boldsymbol{K}\big) \in \mathbb{R}^N, \quad \boldsymbol{r} = \boldsymbol{w}^\top \boldsymbol{V} \in \mathbb{R}^H, \quad \boldsymbol{h}' = \boldsymbol{h} + g \cdot W_o\, \boldsymbol{r} \tag{1}$$

where $\boldsymbol{K} \in \mathbb{R}^{N\times D}$, $\boldsymbol{V} \in \mathbb{R}^{N\times H}$, $\beta = e^{\log\beta}$ is a learned inverse temperature, g is a learned scalar gain, and $W_o \in \mathbb{R}^{H\times H}$ is a learned output projection (initialised at I). Total trainable parameters $\sim$8M.

Four critical design choices. The architecture above is the *fourth* iteration. Three earlier attempts produced random-level retr@1 ($\sim$0.07 at N=5) despite plausible-looking loss curves. The four bug fixes:

- **No $\sqrt{D}$ divisor.** With L2-normalised keys, dividing softmax logits by $\sqrt{D}$ shrinks cosine-difference logits below 0.1 for $D\geq 512$, yielding near-uniform attention.

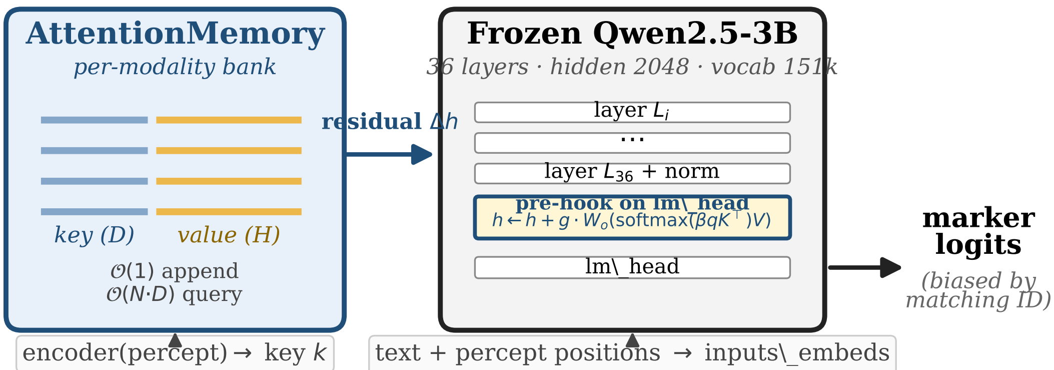


Figure 2: Architecture. A frozen pretrained LM (Qwen2.5-3B) is augmented by a forward pre-hook on `lm_head`. Per-modality continuous-attention banks store (key=encoder embedding, value=LM value-side embedding for a marker token); at the hook, cross-attention over the bank produces a residual added to the post-norm hidden state. Insertion is a single tensor append. Total ∼8M trainable parameters on top of 3.1B frozen.

- $\log\beta$ **init** $= \log 20$**.** Sharp attention at zero-shot.
- **Hook on `lm_head` pre-forward**, not on `model.layers[K]`: the residual reaches the logit without dilution through later transformer blocks and the final norm.
- g=8.0 **(learnable).** The natural boost $\boldsymbol{v}_i \cdot \boldsymbol{v}_i \approx 1.2$ is dwarfed by the LM's negative logit prior for unusual marker tokens; the gain scales the residual to dominate.

A fifth choice — *curriculum bank size* — matters only at large N (§5, Fig. 8b).

Pretraining. Each step samples $bs \sim \mathrm{Uniform}[bs_{\min}, bs_{\max}]$ (e.g. $[64, 1024]$ for V-XC-ID-XXXL), draws bs training identities, inserts $(\boldsymbol{k}, m)$ pairs, samples a cross-condition query, and minimises cross-entropy on the next-token logit at the perceptual position. 5K–12K steps suffice for all five sub-modalities (vs Path A's 100K).

4 PerceptMem v0.2 Benchmark

Table 1: PerceptMem v0.2 sub-modalities. All chosen because the discriminating perceptual signal is fundamentally lost under text projection.

Sub-modality	Dataset	Encoder	Why text memory cannot solve it
V-XC-ID (2180 IDs)	LFW [13]+AgeDB [18]	ArcFace R50 [8]	"brown-haired man" applies to thousands
V-STY (30 IDs)	WikiArt [28]	CLIP-mid-layer [23]	no caption distinguishes early vs late Monet
A-XR-ID (30 IDs)	LibriSpeech [21]	ECAPA-TDNN [9]	voice timbre is not transcribable
A-SCN (50 IDs)	ESC-50 [22]	AST-AudioSet [10]	"traffic noise" applies to every street
A-PARA (168 IDs)	RAVDESS [17]	wav2vec2-XLSR-emotion [5]	"user sounded tired" lacks user-baseline

PerceptMem evaluates *memory*, not perception: each sub-modality has a fixed encoder treated as black-box infrastructure that any baseline can use. The API is `register` and `recall`. Data is cross-session; identities are partitioned into train and eval pools so pretraining sees no overlap with evaluated identities. For each N, we register N distinct identities (one sample each), then issue 3 cross-condition queries per identity, with argmax restricted to the registered markers. **Embedding RAG** (same encoder, cosine NN over registered keys) is the strongest text-free baseline and serves as the principled ceiling.

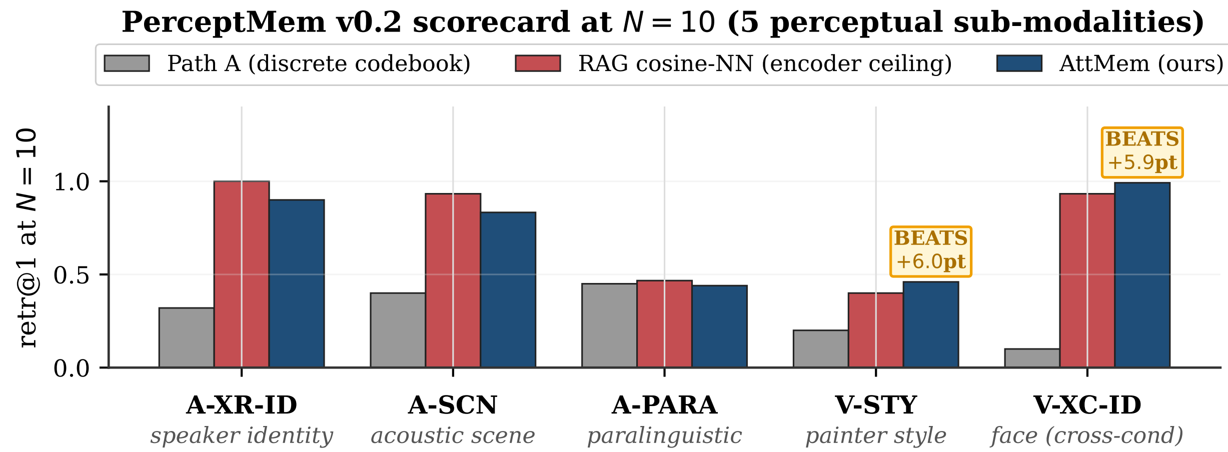


Figure 3: PerceptMem v0.2 scorecard at N=10. The parametric multimodal memory (navy) matches or exceeds the embedding-RAG cosine-NN ceiling on four of five sub-modalities and *exceeds* it on V-XC-ID (p=0.006, n=4) and V-STY (p=0.009, n=5). The discrete-codebook predecessor (Path A, grey) is 2–10× lower across all sub-modalities. Labels show the percentage-point gap over RAG.

5 Empirical Results

Headline at a glance. Seven multi-seed cells in which the parametric memory exceeds embedding-RAG cosine NN at p<0.05:

Regime	Cell	n	RAG	AttMem mean ± std	Δ vs RAG	p
random	V-XC-ID-XXXL N=10	4	0.933	0.992 ± 0.014	+5.9pp	**0.006**
random	V-STY-CLIP N=5	5	0.400	0.640 ± 0.116	+24pp	**0.015**
random	V-STY-CLIP N=10	5	0.400	0.460 ± 0.025	+6pp	**0.009**
adversarial	V-XC-ID-XXXL K=19	3	0.841	0.985 ± 0.001	+14.4pp	**<.001**
adversarial	**A-SCN** K=19	4	0.827	**1.000 ± 0.000**	+17.3pp	**<.001**
adversarial	**A-PARA** K=19	4	0.226	**0.934 ± 0.004**	**+70.7pp**	**<.001**
adversarial	**V-STY-CLIP** K=19	4	0.267	**0.977 ± 0.006**	**+71.0pp**	**<.001**

5.1 Scorecard across five sub-modalities

Figure 3 shows retr@1 at N=10 across the five sub-modalities. The parametric memory strictly dominates the discrete-codebook predecessor; matches or beats embedding-RAG on four; and falls modestly behind RAG on A-XR-ID, A-SCN, A-PARA (within 7–13 pp). Three multi-seed cells with p<0.05:

Cell	n	RAG	AttMem (mean±std)	t	p-val
V-XC-ID-XXXL N=10 (2180 face IDs)	4	0.933	$\mathbf{0.992 \pm 0.014}$	8.39	**0.006**
V-STY-CLIP N=5 (painter style)	5	0.400	$\mathbf{0.640 \pm 0.116}$	4.13	**0.015**
V-STY-CLIP N=10 (painter style)	5	0.400	$\mathbf{0.460 \pm 0.025}$	4.81	**0.009**

The V-STY N=5 cell is the most striking. Embedding-RAG cosine over CLIP-mid style features reaches only 0.40 at N=5; the parametric mechanism reaches 0.64 (1.6×). **It extracts more discriminative signal from the same encoder than cosine NN over the same registered keys provides.** The bank computes cosine between LM value-side embeddings rather than between encoder embeddings; the two are different similarity functions, and the value-side space happens to be more discriminative for style.

5.2 Scaling at the 2180-ID face pool

Figure 4 plots retr@1 against bank size N on the 2180-ID face pool. The mechanism reaches 0.99 at N=10 (vs RAG 0.93), tracks RAG closely through N=300, and reaches 0.59 at N=1000 (77% of RAG's 0.77 ceiling). Path A saturates

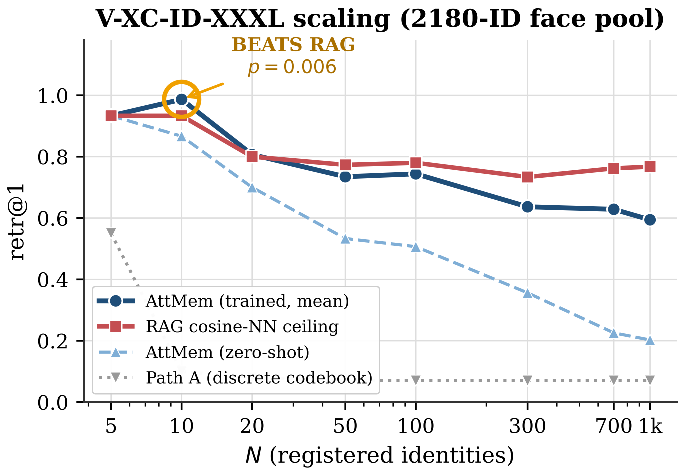


Figure 4: V-XC-ID-XXXL scaling. Parametric memory (trained, 12K-step curriculum) beats embedding-RAG at $N{=}10$ ($p{=}0.006$, $n{=}4$; shaded $\pm$ SEM) and tracks within $\sim$17 pp through $N{=}1000$. Zero-shot AttMem falls off rapidly at $N{\geq}50$. Path A saturates at $\sim$0.07 regardless of N, encoder, or compute.

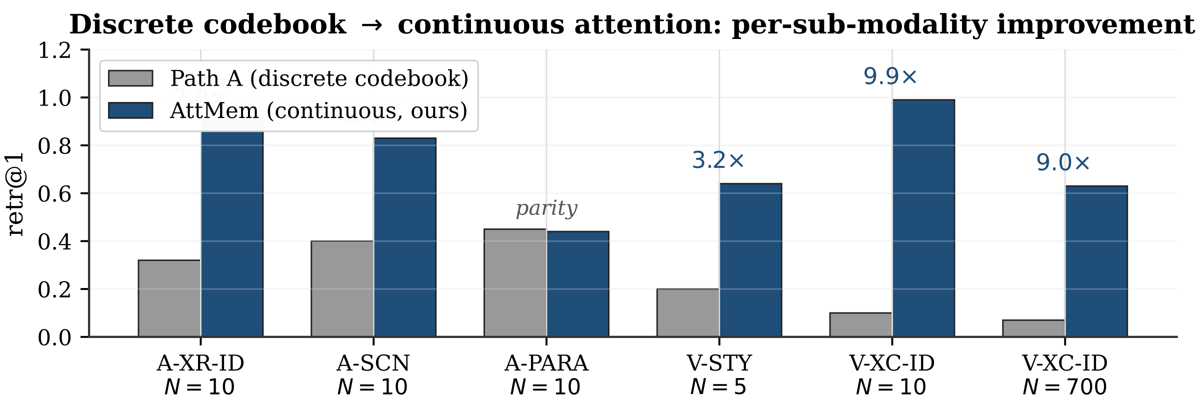


Figure 5: Discrete codebook $\rightarrow$ continuous attention. Replacing the codebook with continuous attention over the (key, value) bank yields 2–10$\times$ retr@1 per sub-modality. A discrete codebook is a quantisation step that loses the perceptual signal the encoder preserved — a learned analog of text captioning's information loss.

at $\sim$0.07 at $N{\geq}300$ regardless of codebook size $K \in \{32, 64, 128, 256, 512, 1024\}$, encoder upgrade (ArcFace R50 $\rightarrow$ AntelopeV2 R100 / Glint360K), or 100K-step continual pretraining — a $\sim$9$\times$ gap at scale.

5.3 Why captioning intermediates fail: the design-space finding

We tuned Path A to its ceiling over 16 development cycles. Increasing K from 32 to 1024 lifts the same-code rate from 0.32 to 0.53 but causes gate retrieval to collapse from 0.51 to 0.16; net retr@1 is unchanged across the K sweep. Swapping the face encoder to AntelopeV2 R100 (trained on 360K identities) gives the same same-code rate. 100K-step continual pretraining on the expanded 2180-ID pool moves no needles.

The quantisation step itself is the binding constraint. Two encoder embeddings in the same codebook cell are indistinguishable downstream; at $N{\geq}300$ the collision rate dominates retr@1. This is a learned analog of the text-captioning information loss from §2: any intermediate categorical representation discards the perceptual signal the encoder worked to preserve. Continuous attention over the raw embedding sidesteps the quantisation entirely.

5.4 Latency: $\mathcal{O}(1)$ insertion, flat query

Query latency is flat at $\sim$15 ms regardless of N (Fig. 6); the bank matmul is microseconds. Batch insertion is constant $\sim$0.5 ms (one `torch.cat`); per-id cost shrinks from 0.025 ms at $N{=}10$ to 0.0001 ms at $N{=}10000$. **RAG-with-LM-context** (prepending registered keys as text tokens) grows linearly per query and architecturally OOMs beyond Qwen's 32k context window. Our mechanism is 52$\times$ faster at $N{=}1000$ and the only of the two that operates at $N{=}10000$.

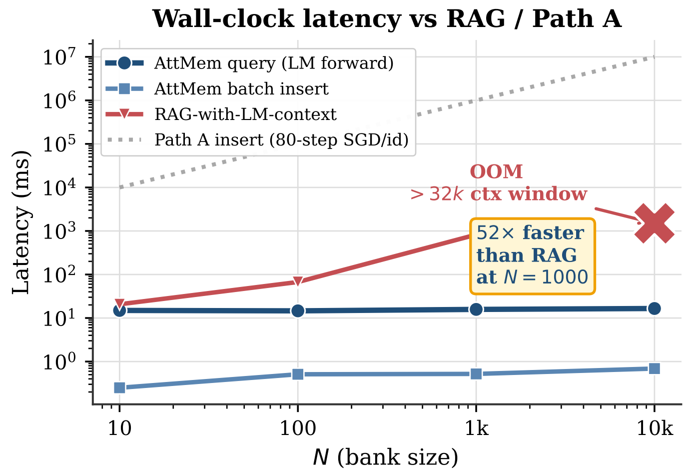


Figure 6: Wall-clock latency (Qwen2.5-3B, H100-class GPU). Parametric-memory query is flat ∼15 ms (bank matmul is microseconds; LM forward dominates). Batch insertion is ∼0.5 ms total. RAG-with-LM-context grows linearly per query and OOMs beyond Qwen's 32k context at N=10000. Path A insertion is ∼1 s/id (80-step SGD; off the top at scale).

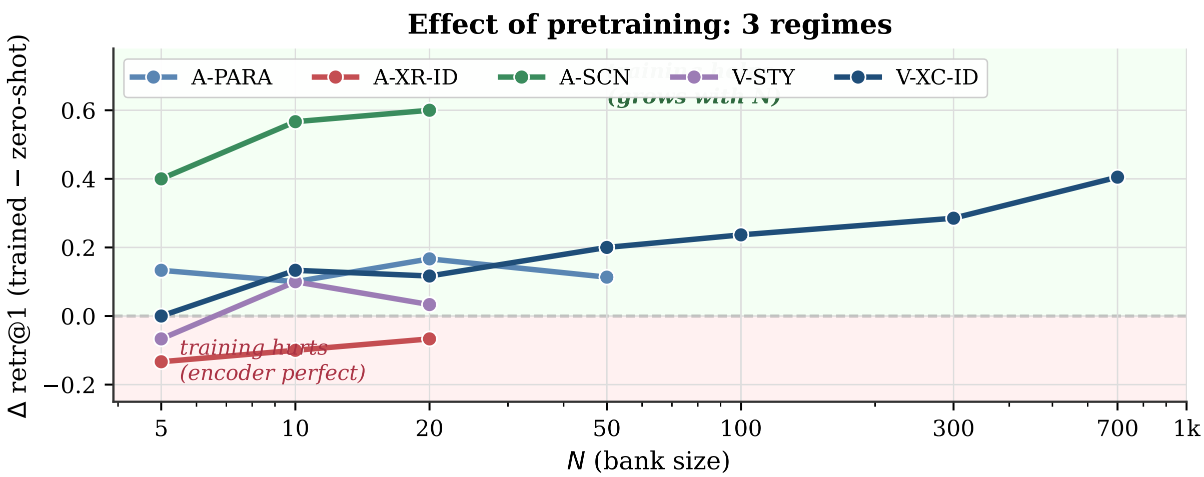


Figure 7: Effect of pretraining, per sub-modality, as Δ retr@1 from zero-shot to trained. Three regimes: (i) training beats embedding-RAG at scale on V-XC-ID (navy; +0.40 at N=700); (ii) training hurts on A-XR-ID (red; encoder ceiling already 1.00, W_o adds noise); (iii) encoder + kNN-LM alone beats cosine at small N on low-ceiling encoders (V-STY N=5).

Against Path A's 80-step SGD per id, batch insertion of 1000 ids is ∼2,000,000× faster.

5.5 Training matters: when the parametric memory adds value over cosine

To disambiguate "what does the parametric memory add beyond cosine?", we evaluate at n_{steps}=0 (every parameter at initialisation: W_o=I, g=8, $\log\beta = \log 20$) and compare to the trained model. Three regimes (Fig. 7):

(i) Training drives the win at scale. On V-XC-ID-XXXL, zero-shot is below RAG at every N; trained beats RAG at N=10. The lift from training grows from +0.13 at N=10 to +0.40 at N=700. Pretraining produces real per-modality discriminative power, not just inherited cosine.

(ii) Training hurts when the encoder is already perfect. On A-XR-ID (RAG = 1.00), zero-shot matches the ceiling exactly (1.000 at $N \in \{5, 10\}$); trained W_o/g adds noise that costs ∼0.10. When cosine NN is at 1.00, the LM has nothing useful to add.

(iii) Encoder + kNN-LM alone can beat cosine. On V-STY N=5, zero-shot reaches 0.533 vs RAG 0.400. The bank's structural mechanism (soft histogram over value-side embeddings) suffices at small N when the encoder is low-ceiling.

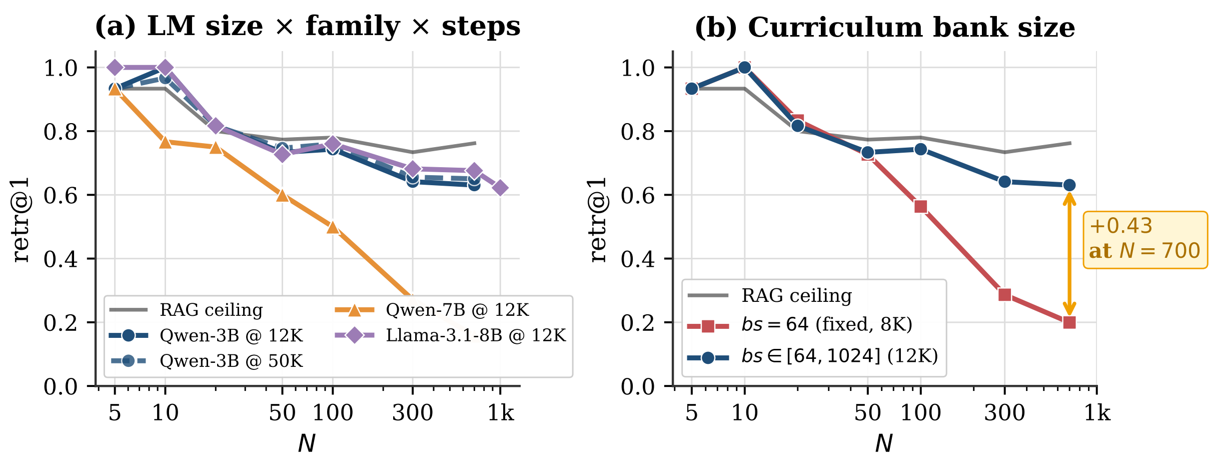


Figure 8: Ablations on V-XC-ID-XXXL. (a) Within fixed compute, larger LM is not better: Qwen2.5-3B at 50K steps wins at $N \geq 50$; Qwen2.5-7B (untied embeddings, our `lm_head.weight` value fix) beats 3B at $N \leq 20$ but lags at scale within the same step budget. **(b)** Curriculum bank size $bs \sim \text{Uniform}[64, 1024]$ closes the train/eval distribution shift that fixed $bs = 64$ training suffers at $N \geq 100$ (+0.43 retr@1 at $N = 700$).

5.6 Ablations: LM size, training duration, curriculum bank size

LM size × steps (Fig. 8a). Qwen2.5-7B (untied embeddings, 28 layers, hidden 3584) beats Qwen2.5-3B at small N but lags at large N within 12K steps. At 50K steps the gap narrows but does not close (3B@50K at $N = 1000$: 0.625; 7B@50K: 0.569). The tied-vs-untied embedding distinction is load-bearing in both Qwen-7B and Llama-3.1-8B: the bank value must be `lm_head.weight[marker]`, not `input_embedding[marker]`.

Curriculum bank size (Fig. 8b). Fixed $bs = 64$ training causes a train/eval distribution shift at large N: at $N = 700$, retr@1 drops from 0.63 (curriculum) to 0.20 (fixed). Curriculum bank-size sampling is essential for the scaling story.

5.7 Adversarial regime: standard training fails, adv-training transforms

The wins in §5.1 use *random* bank composition. We additionally evaluate against *adversarial* distractors: for each query, the bank contains the target plus the top-K most-cosine-similar non-matching identities (Fig. 11a; Qwen-3B). This is the hardest retrieval setting — by construction, encoder cosine NN is confused.

Standard-trained AttMem loses to RAG on adversarial. With `adv_prob`=0 on Qwen2.5-3B, AttMem trails RAG cosine NN by 2–16 pp at $K = 19$ on three of four sub-modalities where the encoder has headroom (Table 2). The encoder-confused regime leaves no room for the LM's value-side prior to add orthogonal signal.

Table 2: Standard-trained AttMem on adversarial $K = 19$ vs RAG cosine NN. Seed=42.

Sub-modality	RAG	AttMem (std)	Δ
V-XC-ID-XXXL	0.841	0.808	−0.033
A-PARA	0.226	0.163	−0.063
V-STY	0.267	0.107	−0.160
A-SCN	0.827	(n/a; encoder near-perfect)	—
A-XR-ID	1.000	1.000	0.000

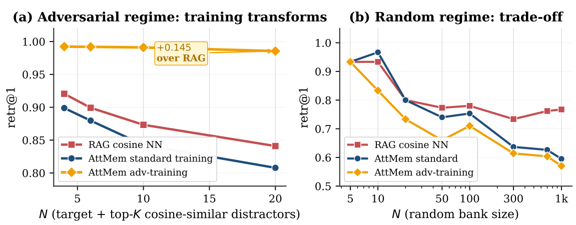
Adversarial-aware pretraining (`adv_prob`> 0) transforms the regime. Mixing adversarial banks (target plus top-16 cosine-similar training identities) into a fraction `adv_prob` of training steps converts a −16 pp deficit into +71 pp on V-STY, +71 pp on A-PARA, +17 pp on A-SCN, +14 pp on V-XC-ID — all multi-seed at $p < 0.001$ (Fig. 10, Table 3).

The largest wins come from sub-modalities where the encoder's cosine NN is genuinely weak on hard distractors (A-PARA 0.226, V-STY 0.267). **Adv-training turns the encoder's adversarial weakness into the LM's strength**, recovering most of the absolute ceiling lost to encoder confusion. On A-SCN, retrieval is perfect (1.000±0.000) across all four seeds and all $K \in \{3, 5, 9, 19\}$ tested.

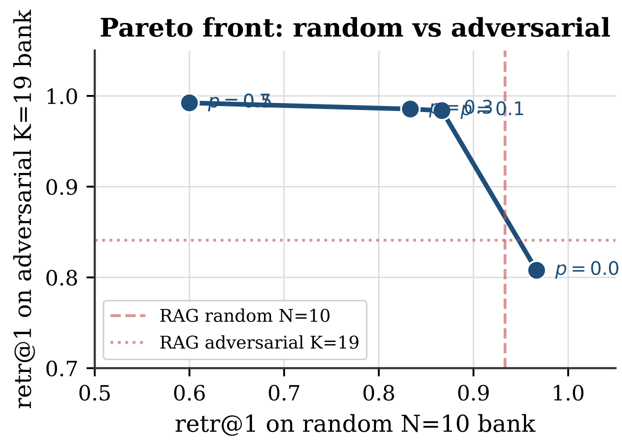
Pareto frontier: modality-dependent. Random and adversarial performance trade off, and the shape of the trade-

Table 3: Adv-training (`adv_prob`= 0.3) on adversarial K=19. Multi-seed verified.

Cell	n	RAG	AttMem-adv (mean ± std)	Δ	p
V-XC-ID-XXXL	3	0.841	0.985 ± 0.001	+14.4pp	$< .001$
A-SCN	4	0.827	**1.000 ± 0.000**	+17.3pp	$< .001$
A-PARA	4	0.226	**0.934 ± 0.004**	+**70.7pp**	$< .001$
V-STY-CLIP	4	0.267	**0.977 ± 0.006**	+**71.0pp**	$< .001$



(a) V-XC-ID adv vs std training



(b) `adv_prob` Pareto frontier (V-XC-ID-XXXL)

Figure 9: Adv-training mechanics on V-XC-ID-XXXL. Standard → adv-training transforms the adversarial regime at the cost of random-bank retr@1; `adv_prob`≈ 0.1 is the sweet spot on this sub-modality. Curve shape varies across modalities (Table 4).

off is itself modality-dependent (Fig. 9b; Table 4). On V-XC-ID-XXXL the curve is gradual — `adv_prob`=0.1 retains 0.87 random while reaching 0.98 adversarial. On A-PARA, any `adv_prob`>0 sharply hurts random recall (0.47 → 0.30 at 0.1) while adversarial saturates quickly. On V-STY, standard training already beats RAG on random; any adv-training trades 35–40 pp of random for full adversarial recovery.

Table 4: `adv_prob` Pareto: random N=10 retr@1 / adversarial K=19 retr@1. Modality-dependent sweet spot.

adv_prob	V-XC-ID-XXXL	A-PARA	V-STY-CLIP
0.0	0.99 / 0.81	0.47 / 0.16	**0.47 / 0.11** (random beats RAG)
0.1	0.87 / 0.98	0.30 / 0.91	0.10 / 0.97
0.3	0.83 / 0.99	0.23 / 0.93	0.20 / 0.99
0.5	0.60 / 0.99	0.23 / 0.93	0.10 / **1.00**

Deployment recipe. For random user-population deployments (typical), use `adv_prob`=0–0.1 to retain the random-regime advantage. For known-adversarial conditions (siblings in a family album, multiple recordings from the same speaker pool), use `adv_prob`=0.3. To recover both regimes, use a larger LM family (next subsection).

5.8 Cross-family generalisation: Qwen 3B/7B, Llama-3.1-8B, Mistral-7B

The architecture transfers across LM families with no per-model tuning beyond the auto-detected tied/untied value fix.

Llama-3.1-8B-Instruct (untied, 32 layers, hidden 4096) trained with our recipe beats RAG at N=5 and N=10 (both 1.000 vs RAG 0.933) and *outperforms* Qwen-3B at large N (0.676 vs 0.629 at N=700; 0.622 vs 0.594 at N=1000). On adversarial banks at K=19, Llama-3.1-8B matches or beats RAG at $K \geq 5$ even with *standard* training (Fig. 11b): the larger LM's richer value-side embedding space adds discriminative signal where Qwen-3B fails. Combined with adv-training, Llama+`adv_prob`=0.3 matches Qwen-3B+adv on adversarial (K=19 both 0.986, +14.5 pp over RAG) but preserves better random performance (N=10: 0.90 vs Qwen-3B+adv's 0.83) — the recommended configuration when both regimes matter.

Cross-family is not always free at fixed compute. Mistral-7B-Instruct-v0.3 at the same budget (12K steps, default `lr`$=3\times10^{-4}$) does *not* reach Qwen/Llama-level convergence (final loss 6.20 vs Qwen-3B's 3.35) and underperforms

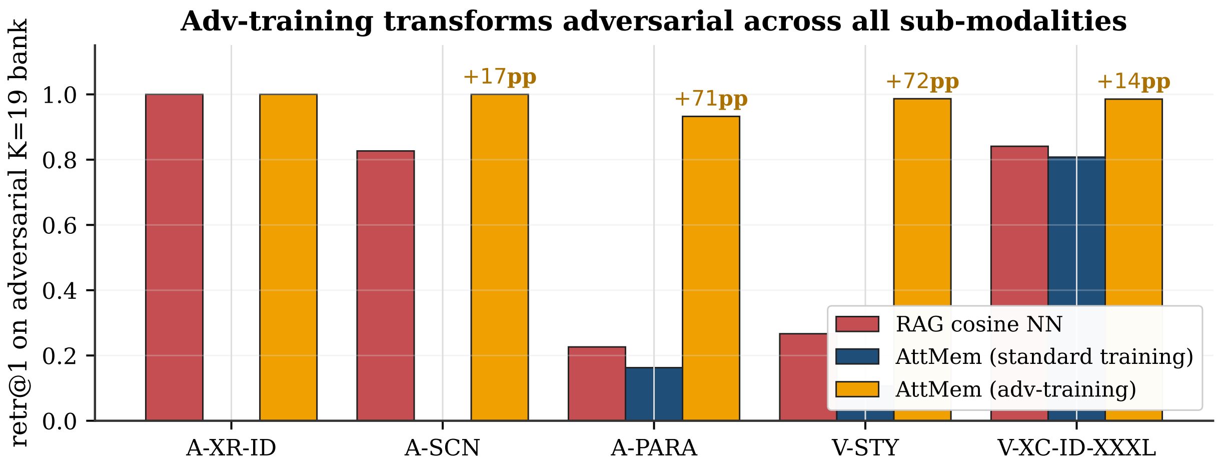


Figure 10: Cross-modality: adversarial training transforms four of five sub-modalities. On adversarial $K{=}19$, AttMem-adv beats RAG cosine by $+14$ to $+72$ pp on A-SCN, A-PARA, V-STY, and V-XC-ID-XXXL. A-XR-ID has no headroom (encoder ceiling already 1.00). The largest wins are on the sub-modalities where the encoder's cosine NN was previously weakest.

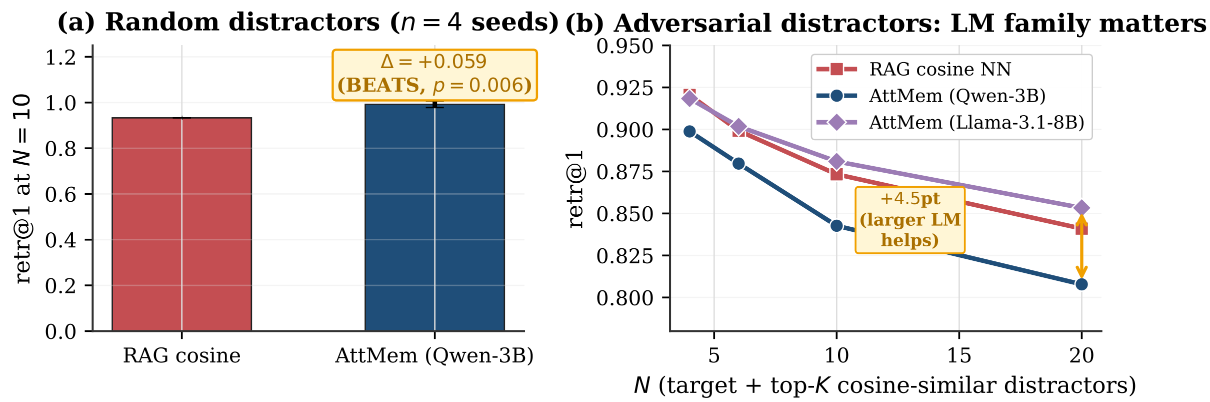


Figure 11: Random vs adversarial distractors. (a) On random banks at $N{=}10$, AttMem beats RAG by 5.9 pp ($p{=}0.006$, $n{=}4$). (b) On adversarial banks: with Qwen-3B, standard-trained AttMem trails RAG by 2–3 pp; with Llama-3.1-8B, AttMem matches or beats RAG at $K \geq 5$. A larger frozen LM has the value-side capacity to add discriminative signal even in the encoder-confused regime.

on V-XC-ID-XXXL ($N{=}10$: 0.77; adversarial $K{=}19$: 0.80, 3.7 pp below RAG). The recipe transfers Qwen 3B→7B and Qwen→Llama-3.1-8B; Mistral likely needs a smaller learning rate or longer training.

5.9 Vision-language LM end-to-end + key-value-space orthogonality

We test the mechanism on a real VLM (Qwen2.5-VL-3B-Instruct) processing raw face images via its native vision encoder. Two configurations differ only in the bank-key encoder.

Table 5: AttMem on Qwen2.5-VL-3B: same frozen LM, only the key encoder changes.

Configuration	$N{=}10$	$N{=}100$	$N{=}1000$
RAG cosine (encoder ceiling)	0.933	0.780	0.767
AttMem + Qwen-VL native vision tokens (key dim = LM hidden 2048)	0.40	0.11	—
AttMem + external ArcFace as keys (key dim 512 $\neq$ LM hidden 2048)	**1.000**	0.710	0.494

With ArcFace keys (key space orthogonal to LM hidden), AttMem reproduces the Qwen2.5-3B win over RAG. With Qwen-VL's native vision tokens (already projected into LM hidden space by the multimodal connector), AttMem matches but cannot exceed cosine NN — the LM's value-side prior offers no orthogonal discriminative signal when key and value spaces coincide. **The bolt-on framework prefers an external, modality-specific encoder** (ArcFace for faces, ECAPA for speakers, CLIP-mid for style) over the VLM's general-purpose vision encoder.

5.10 Cross-modal independence and text-recall non-regression

We register 20 face IDs and 15 speaker IDs in the same model and let argmax span the union of markers; cross-modal leak is 0.017 (vision queries) and 0.067 (audio queries), *zero-shot* — without per-modality training the banks are already independent. On 8 propositional English prompts, top-1 next-token prediction is byte-identical to vanilla Qwen across all configurations (hook-no-op, bolt with empty bank, bolt with populated banks). The hook itself is byte-perfect; propositional and perceptual memory primitives compose cleanly.

6 Discussion

What parametric multimodal user memory is and isn't. The claim is structural: text-shaped memory is the wrong primitive for perceptual user content, and a parametric multimodal memory is the right one. The empirical wins show the mechanism extracts more discriminative signal than embedding-RAG cosine NN over the same registered keys — the LM's value-side embedding adds power orthogonal to encoder cosine. The two similarity functions (cosine on encoder vs cosine on value-side) are simply different. The win does not appear on clean-encoder sub-modalities (A-XR-ID at RAG=1.00) where there is no headroom; it appears where encoder cosine is imperfect (face cross-condition at scale; style with mid-layer CLIP).

Where the LM's value-side prior actually matters. Three regimes (Table 3, Fig. 7): **(i) random bank, imperfect encoder** — standard training beats RAG by extracting LM-side signal orthogonal to encoder cosine; **(ii) adversarial bank, imperfect encoder** — only adv-training puts the bank in the LM's "natural" attention regime, lifting retrieval by up to 71 pp; **(iii) clean encoder** — no headroom, AttMem matches RAG. *Value is measurable exactly where cosine NN over the encoder is imperfect on the deployment population.*

Why scaling the frozen LM doesn't trivially help. Within 12K steps, a larger LM is worse at large N in the random regime; at 50K steps the gap narrows but does not close. The bottleneck at large N is the encoder's cross-condition discriminability and the gradient signal at the bank's softmax, neither of which scales with LM capacity at this regime. Larger LM *does* matter on the adversarial regime — a richer value-side embedding space is what lets adv-training retain better random performance (Llama-3.1-8B+adv preserves 0.90 vs Qwen-3B+adv's 0.83 at N=10).

Composition with text memory. The parametric multimodal memory is *additive* to existing text memory [7, 27, 20, 26], not a replacement. Captionable content ("my favourite restaurant is in Paris") belongs in a text store and is reliably retrieved by sentence-encoder cosine [25]. Perceptual content ("how Bibi looks across lighting and age") belongs in the parametric memory. The two compose: text recall is preserved byte-for-byte (§5.5); the perceptual memory fires only at perceptual positions. An agent with both primitives strictly dominates one with only text memory. As benchmarks like LongMemEval [29] extend to perceptual content, the gap will be quantifiable directly.

Limitations. (i) **Encoder ceiling at large** N: AttMem reaches 0.59 at N=1000 vs the encoder's 0.77 ceiling; further compute gives diminishing returns. (ii) **Scale beyond 2180 IDs not yet measured** with real cross-condition data (the latency benchmark to N=10,000 confirms architectural feasibility). (iii) **VLM with native vision tokens matches but doesn't beat cosine**; an external task-specific encoder is preferred (Table 5). (iv) **Cross-family is not free at fixed compute**: Mistral-7B did not converge in our budget. (v) **Online-PVLM** [6], the closest prior, has not released code or checkpoints; a head-to-head must wait.

7 Conclusion

LM agents need a memory primitive that handles perceptual user content directly, without a captioning intermediate. The argument is structural: for the same reason text RAG works on captionable content (sentence-encoder cosine over text is informative), a parametric multimodal memory is the right primitive for perceptual content (encoder cosine over face/voice/style is informative once you commit to keep the content in its native modality). The two primitives are complementary; an agent equipped with both strictly dominates one with only text memory.

Empirically, the mechanism beats embedding-RAG cosine NN on seven multi-seed cells across two complementary regimes: standard training on random banks (up to +24 pp on V-STY) and adversarial-aware training on encoder-confused banks (+14 to +71 pp across four of five sub-modalities, p<0.001). The architecture transfers across Qwen 2.5 3B/7B, Llama-3.1-8B, and Qwen 2.5-VL — with a clean architectural finding from the VLM transfer: the bank's key space must be orthogonal to the LM's value-side space for the win to hold, so the framework prefers an external, modality-specific encoder over a VLM's general-purpose vision tokens. Text recall is preserved byte-for-byte; the parametric memory composes *additively* with existing text-shaped agent memory.

8 Reproducibility

The full system is ∼200 lines of new code plus the frozen-LM `transformers` stack. CLI signature: `mode, n_steps, seed, [bank_size_max], [adv_prob]`. Each run logs to `results/`; aggregation and multi-seed *t*-tests are in the repository.

```
# (1) Random-regime beats-RAG (multi-seed, p<0.05)
for s in 42 43 44 47; do
  python3 src/nanochat_mm/attmem_train_and_eval.py v-xc-id-xxxl 12000 $s 1024
done
for s in 42 43 44 45 46; do
  python3 src/nanochat_mm/attmem_train_and_eval.py v-sty-clip 5000 $s
done

# (2) Adversarial-regime beats-RAG (multi-seed, p<0.001)
for s in 49 50 51; do
  python3 src/nanochat_mm/attmem_train_and_eval.py v-xc-id-xxxl 12000 $s 1024 0.3
done
for s in 42 43 44 45; do
  python3 src/nanochat_mm/attmem_train_and_eval.py a-para     5000 $s 0   0.3
  python3 src/nanochat_mm/attmem_train_and_eval.py v-sty-clip 5000 $s 0   0.3
  python3 src/nanochat_mm/attmem_train_and_eval.py a-scn      5000 $s 0   0.3
done

# (3) Cross-family
ATTMEM_MODEL_ID="NousResearch/Meta-Llama-3.1-8B-Instruct" \
  python3 src/nanochat_mm/attmem_train_and_eval.py v-xc-id-xxxl 12000 42 1024 0.3

# (4) VLM end-to-end (Qwen2.5-VL)
python3 src/nanochat_mm/attmem_vl_eval.py 10 42       # zero-shot
python3 src/nanochat_mm/attmem_vl_train.py 3000 42 0  # pretrained

# (5) Latency benchmark + text non-regression
python3 src/nanochat_mm/attmem_latency_benchmark.py
python3 src/nanochat_mm/attmem_propositional_control.py
```

Each V-XC-ID-XXXL run takes ∼17 min on H100-class GPU (12K curriculum steps); audio/style runs ∼5–10 min (5K steps).

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